

Recap

What we've learnt so far

Quick Recap

1. Visual representation learning: pretraining + evaluation
2. Generative vs. pretext task vs. joint-embedding methods
3. Joint-embedding methods:
 1. Invariant to data augmentation
 2. Prevent trivial solution
4. Contrastive methods
 1. Hard negative mining
 2. Large negative sample pool (SimCLR vs. MoCo)

Non-contrastive methods

Prevent trivial solution without negative samples

The Disadvantages of Contrastive Methods

Contrastive methods require techniques such as:

1. weight sharing between the branches
2. batch normalization
3. feature-wise normalization
4. output quantization
5. stop gradient
6. memory banks
7. ...

Non-contrastive methods and information theory

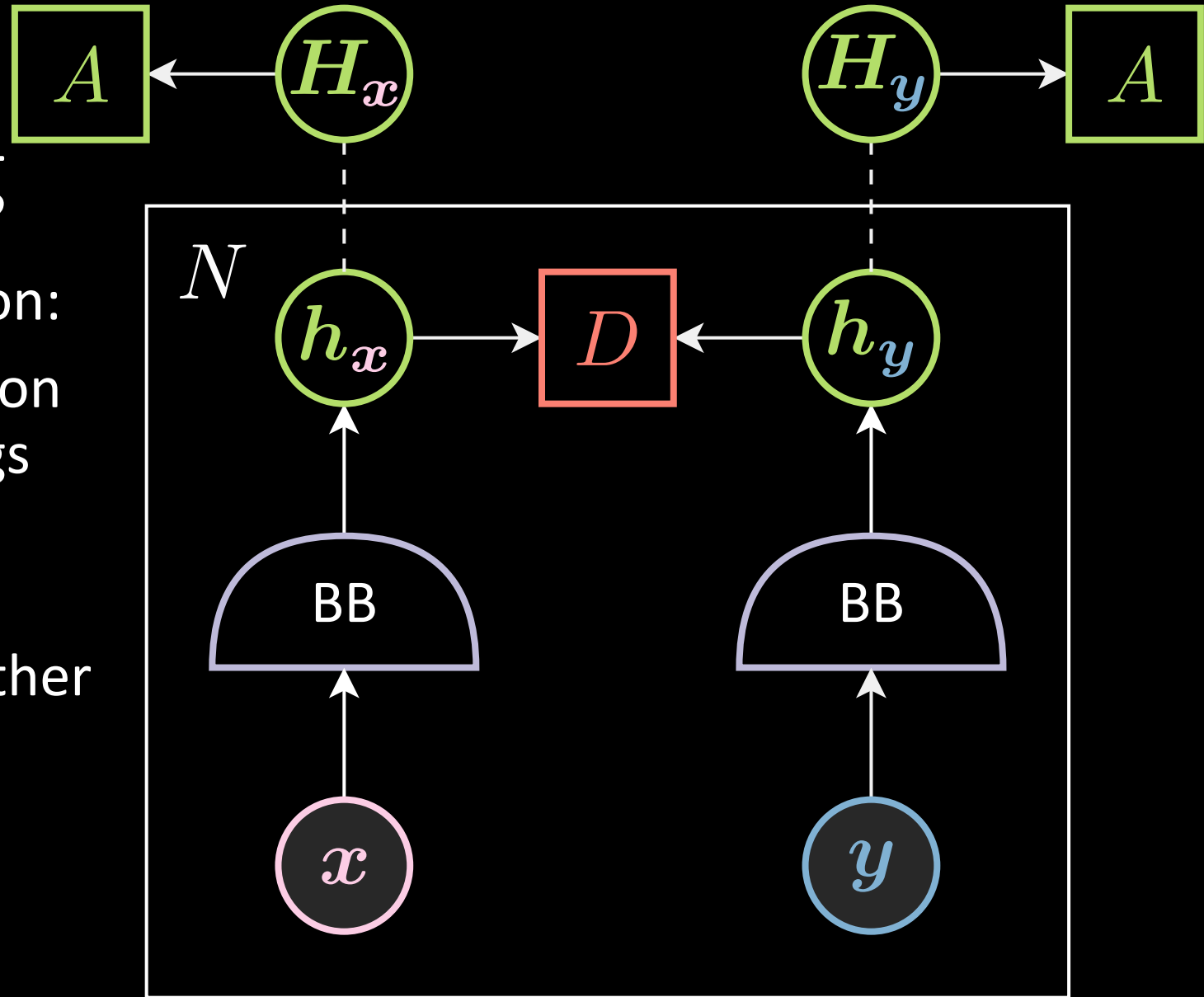
Most of non-contrastive methods

- based on information theory
 - Redundancy reduction (Barlow Twins)
 - Information maximization (VICReg)
- Don't require special architectures

Non-Contrastive Methods – VICReg

The Information maximization:

1. to maximize the information content of the embeddings
2. produce embedding variables that are decorrelated from each other
3. prevent an informational collapse in which the variables carry redundant information



Non-Contrastive Methods – VICReg

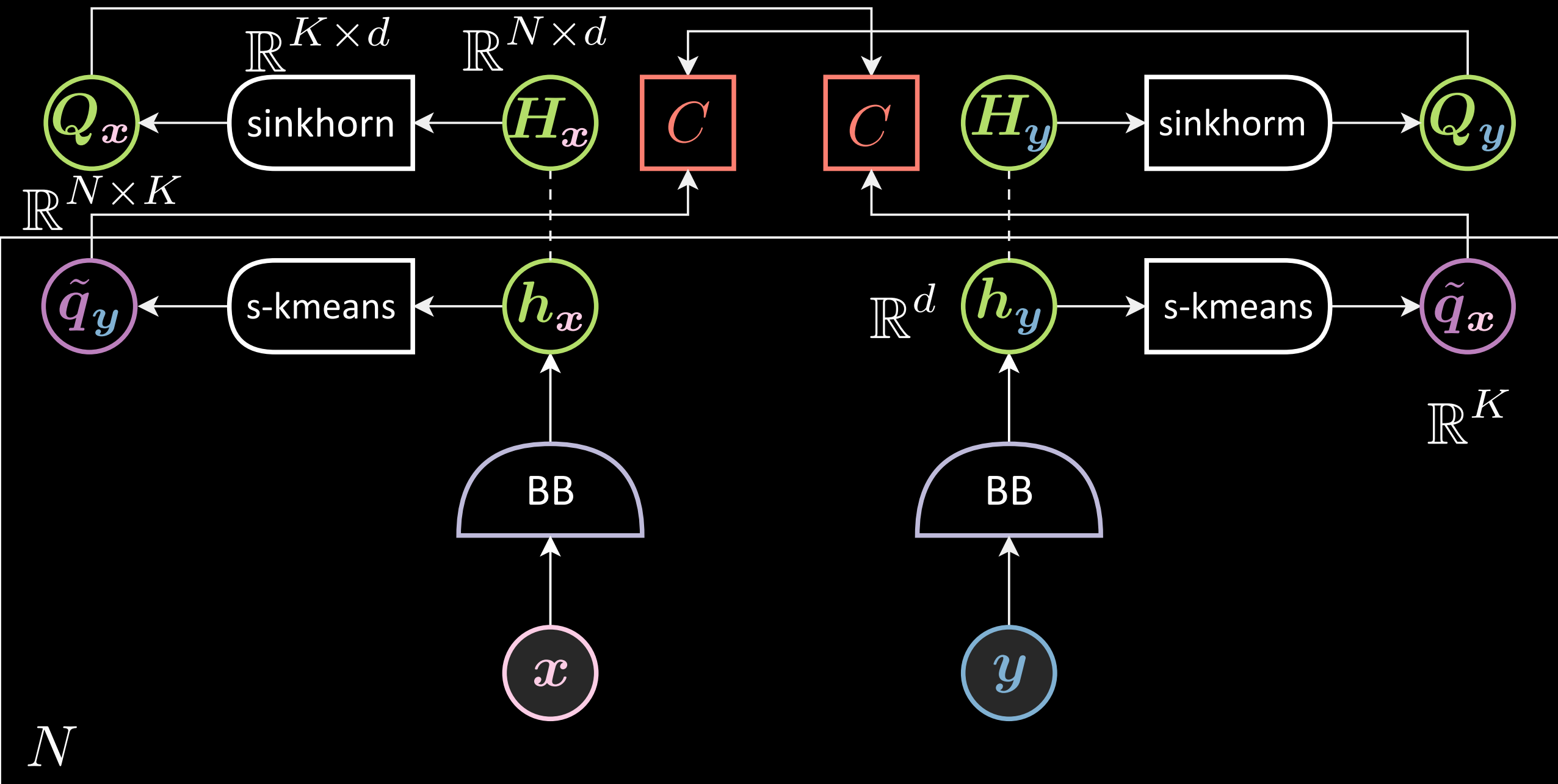
The loss function is pushing

1. the positive pairs closer ($\mathbf{h}_x^i, \mathbf{h}_y^i$)
2. the variance of the embeddings large
3. The covariance of the embeddings small

$$\begin{aligned} L(\mathbf{w}, \mathbf{x}, \mathbf{y}) &= \|\mathbf{h}_x - \mathbf{h}_y\|^2 \\ &+ \frac{1}{d} \left[\sum_i^d (\gamma - \mathbf{x} \mathbf{C}_{ii})^+ + (\gamma - \mathbf{y} \mathbf{C}_{ii})^+ \right] \\ &+ \frac{1}{d} \left[\sum_i^d \sum_{j \neq i}^d \mathbf{x} \mathbf{C}_{ij}^2 + \mathbf{y} \mathbf{C}_{ij}^2 \right] \end{aligned} \quad \mathbf{C} = \frac{1}{N} \mathbf{H}^\top \mathbf{H}.$$

Clustering Methods - SwAV

Prevent trivial solution by quantizing the embedding space



SwAV (II)

$$Q_x = \text{sinkhorn}_W(\mathbf{H}_x) \in \mathbb{R}^{N \times K}$$

$$Q_x = [\mathbf{q}_x^1, \dots, \mathbf{q}_x^N]^\top$$

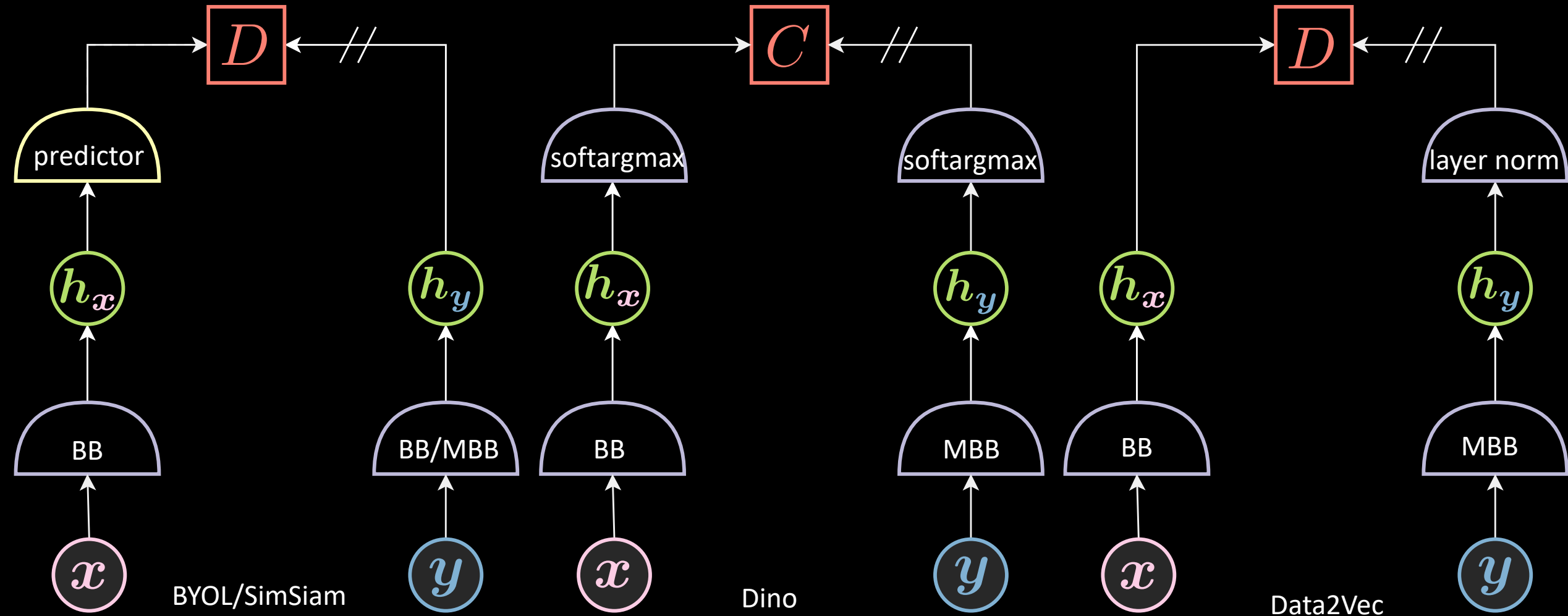
$$\mathbf{W} \in \mathbb{R}^{K \times d} : \text{dictionary}$$

$$\tilde{\mathbf{q}}_{\textcircled{x}} = \text{softargmax}_\beta(\mathbf{W} \mathbf{h}_{\textcircled{y}}) \in \mathbb{R}^K$$

$$F(x, y) = C(\mathbf{q}_x, \tilde{\mathbf{q}}_x) + C(\mathbf{q}_y, \tilde{\mathbf{q}}_y)$$

Other methods

Other Methods – BYOL, SimSiam, Dino, Data2Vec



Grill, Strub, Altché, Tallec, Richemond, Buchatskaya, ... & Valko (2020). Bootstrap your own latent-a new approach to self-supervised learning.

Chen, X., & He, K. (2021). Exploring simple siamese representation learning.

Caron, Touvron, Misra, Jégou, Mairal, Bojanowski & Joulin (2021). Emerging properties in self-supervised vision transformers.

Baevski, Hsu, Xu, Babu, Gu & Auli (2022). Data2vec: A general framework for self-supervised learning in speech, vision and language.

Data augmentation and network architecture

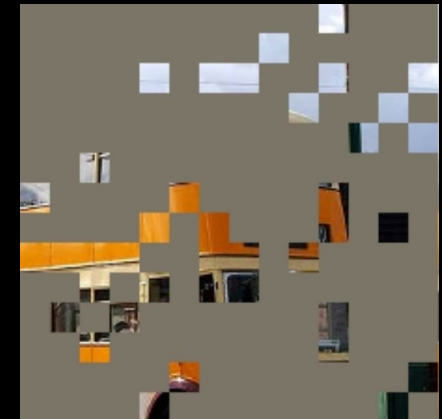
Data augmentation

The SimCLR/BYOL data augmentation:

1. Random Crop (the most critical one)
2. Flip
3. Color Jitter
4. Gaussian Blur



For traditional augmentation to
masking augmentation



Network architecture

It is always better to add a two/three-layer projector/expander

Even without memory bank, momentum encoder usually helps the performance of the downstream tasks, especially with weak data augmentation

